The Legacy of the Descartes-Fermat Feud on Analytic Geometry and Calculus

The discovery of the tangent line of a curve is one of the foundational problems in mathematics, dating back to ancient Greece. This problem has been studied and developed by several mathematicians, including Euclid, Apollonius, Fermat, and Descartes. In the 17th century, Fermat and Descartes developed their own methods for computing the tangent line of a curve at any specific point. Although both mathematicians came up with a method that worked, they disagreed on the other's method, leading to a feud that helped form the basic intuitions of calculus that would be picked up by Newton and Leibniz for the foundations of calculus. This paper aims to analyze the feud between Fermat and Descartes over the discovery of the tangent line of a curve.

The earliest developments of the tangent line problem come from the works of Euclid and Apollonius. These ancient mathematicians had similar definitions of tangent lines in their individual works, "Elements" and "Conics," respectively. Although their works spoke of tangents to circles, they paved the way for many mathematical ideas going forward and would be picked up by Fermat and Descartes in the 17th century.

The tangent line problem involves finding the slope of a curve at a specific point. This problem was studied by Fermat in the 17th century, who developed a method for finding the slope of the tangent line using the limit of the difference quotient. Simultaneously, Descartes formalized ideas that related geometric shapes and algebraic techniques with the development of the Cartesian coordinate system in his work, "La Géométrie," which united algebra and geometry into a unified subject named Analytic Geometry. With this unification came Descartes' method for calculating the tangent line with normal lines, circles, and algebraic methods.

Importance of the Tangent Line Problem

The tangent line problem is important in a variety of fields, including physics, engineering, and economics. For example, in physics, the slope of the tangent line of a displacement graph represents the velocity of an object at a specific point in time. It also extends to higher order principles such as partial derivatives.

Descartes' Method of Computing the Tangent Line of a Curve with the Method of Normals

Rene Descartes developed a method for finding the tangent line to a curve at a given point, known as the method of normals. The method involves finding the line perpendicular to the tangent line at the point of interest, called the normal line. This tangent equation would also be a tangent to some arbitrary circle with radius with length of the original point to the x-axis in a

straight line. Then with algebraic manipulations of these found values, eventually the slope of the normal line was found, which happened to be the negative reciprocal of the original line.

This method was plausible but was very complex for curves represented by polynomials. Also, this method required the curve in question to be explicitly defined. In other words, implicitly defined functions were seemingly not possible to have a method for finding a tangent line at a given point.

Fermat's Method of Computing the Tangent Line of a Curve via Adequality

Pierre de Fermat developed a method for finding the tangent line to a curve using the concept of adequality, which is a notion of equality between infinitesimal quantities. Fermat's approach involves comparing two infinitesimal quantities associated with the curve at the point of interest, one corresponding to a small change in the value of x and the other corresponding to the corresponding change in the value of y. By setting these two infinitesimal quantities equal to each other, Fermat was able to find the slope of the tangent line at the

This paper aims to analyze the feud between Fermat and Descartes over their different methods for finding the tangent line of a curve. The feud may have been a result of a disagreement over the formalization of calculus, but it ultimately helped lay the foundation for calculus as we know it today.

Key Point 4: This section will explain the disagreements between Descartes and Fermat over their methods.
Descartes and Fermat had different approaches to finding the tangent line of a curve. While Descartes used the method of normals, Fermat used the concept of adequality. Fermat’s approach was based on infinitesimals, which were viewed as unrigorous by Descartes, who preferred a more algebraic and geometric approach.

Descartes viewed Fermat's method as unrigorous, as Fermat's approach relied on the concept of infinitesimals, which were not yet formally defined at the time. Descartes preferred a more algebraic and geometric approach and saw his method of normals as more rigorous.

Furthermore, there were disputes between the two mathematicians regarding the scope of their methods. Descartes' method of normals only worked for curves that were explicitly defined, while Fermat's method worked for both explicitly and implicitly defined curves. However, Fermat's method was not always easy to apply in practice.

Key Point 5: This section will explain the legacy of the feud between Fermat and Descartes on the development of calculus.
Despite the disagreements between Descartes and Fermat, their work helped lay the foundation for calculus. Both methods involved finding the tangent line of a curve, which is a fundamental

concept in calculus. Descartes' method of normals contributed to the development of the calculus of variations, while Fermat's method helped to develop differential calculus.

Fermat's use of infinitesimals paved the way for the development of differential calculus, which was later formalized by Leibniz. Meanwhile, Descartes' method of normals helped lay the foundation for the calculus of variations, which was later developed by Euler.

Conclusion:
In conclusion, the feud between Fermat and Descartes over their methods for finding the tangent line of a curve was a result of their different approaches to mathematics. While Descartes favored a more algebraic and geometric approach, Fermat relied on the concept of infinitesimals, which were not yet rigorously defined. Despite their disagreements, their work helped lay the foundation for calculus and contributed to the development of differential calculus and the calculus of variations. Their feud ultimately helped shape the basic intuitions of calculus that would be picked up by Newton and Leibniz for the foundations of Calculus.